VR Graffiti

Learning the motion behind the mark.

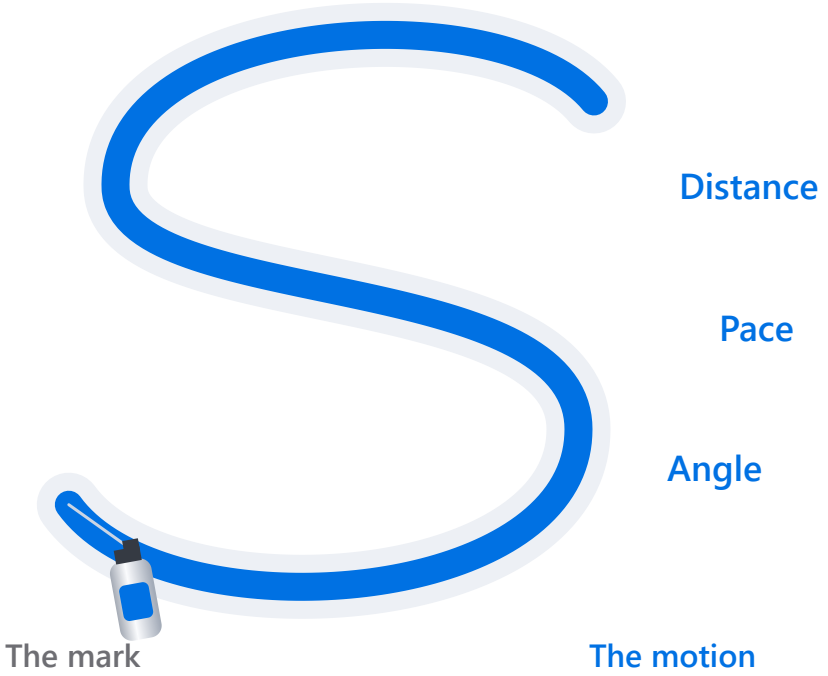


Guided lettering. Recorded gestures. A new creative medium.

Smart Chen · Solo XR concept · Interaction visualization

A finished letter hides the movement.

For beginners learning graffiti and expressive lettering.



Space to practice

Permitted walls.
Paint and ventilation.

Technique to learn

Distance, angle, speed and stroke order.

18 studies in a review of observational motor learning.
VR graffiti remains untested.

Han et al., 2022. Krylon, How to Spray Paint. Beginner needs are design assumptions to validate.

A learning sequence built around the gesture

Existing approach	Established capabilities	This proposal explores
Kingspray	VR spray, replay, projection and interactive tutorials*	Motion guidance with fading cues and an unguided assessment
Open Brush	Expressive painting in three-dimensional space	Recorded gestures that connect learning and temporal artwork
Video + practice	Artist demonstrations and real material feedback	Tracked motion with feedback relative to the virtual wall

An integrated lesson, with learning outcomes still to test.

Kingspray, official PC description. Open Brush. *PC feature description. Capabilities may differ across platforms.

Four styles. Four real references.

Real lettering above. Animated construction below.

Handstyle



YANK tag

Crazyank · CC BY-SA 3.0



YANK study

A signature with rhythm

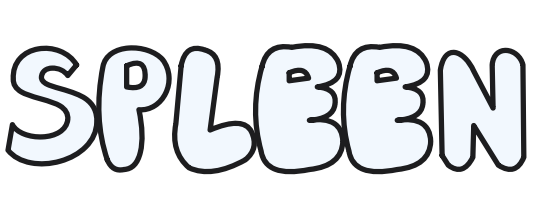
Curriculum direction

Bubble / Throw-up



SPLEEN, NSW

Notcharizard · CC BY-SA 4.0



SPLEEN study

Rounded, inflated contours

Semester lesson: Bubble S

Block / Blockbuster



1UP, Berlin

Singlespeedfahrer · CC0



1UP study

Broad fill and sharp edges

Curriculum direction

Wildstyle



Germany, wall detail

KOchstudiO · CC BY 3.0



FLOW study

Skeleton, overlaps, extensions

Future showcase

Cropped photo details; select an image to view the original. Animated studies illustrate construction, not the artist's recorded stroke order.

Three tools. Different control.

One studio, a learning wall and room to create.



Montana BLACK 400 ml

Spray can



Distance shapes the footprint



MTN Street Paint 15 mm

Paint marker



Contact gives a crisp edge



Purdy Mini Roller System

Paint roller



Broad passes build coverage

Controllers approximate the motion. Physical pressure, friction and tool weight remain absent.

Product photos: Montana Cans product photography. MTN / Spray Planet product photography. Purdy product photography. Motion illustrations show proposed virtual behavior.

A first lesson. A gradual handoff.

The guide steps back as the learner takes control.



Trace with spatial cues that gradually fade

Observe

See tool pose and pace.

Trace

Follow spatial motion cues.

Reproduce

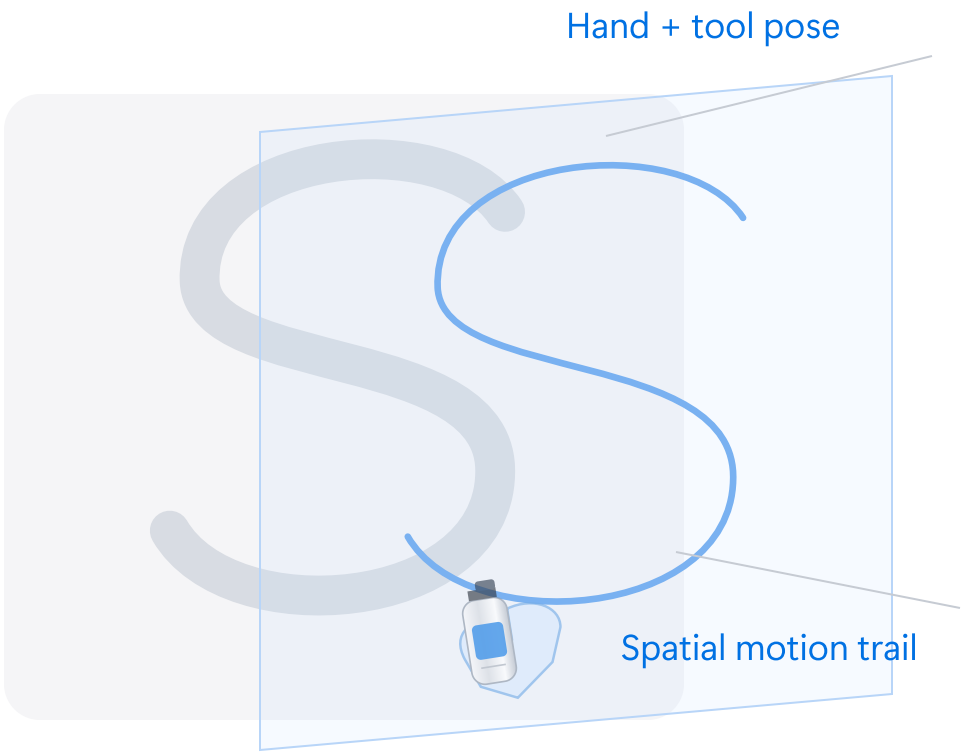
Paint without the overlay.

Remix

Change the letter deliberately.

Concept visualization. Support can be requested again. Remix has no single correct trajectory.

Guidance that steps back



Observational learning

See stroke order and timing before the first attempt.

Scaffolding and fading

Full demonstration, faint trail, then independent movement.

Feedback and reflection

Distance cues while practicing. A trajectory comparison afterward.

Embodied interaction

Tracked tool motion makes the mark in the same space as the guidance.

Han et al., 2022. Wulf et al., 2010. Yu et al., CHI 2024. Levac et al., 2019. Design hypotheses, not established outcomes for this application.

An unguided test of learning

Does ghost guidance help beginners acquire basic VR graffiti technique?

Optional exploratory pilot

A

Visual demonstration and a static target

B

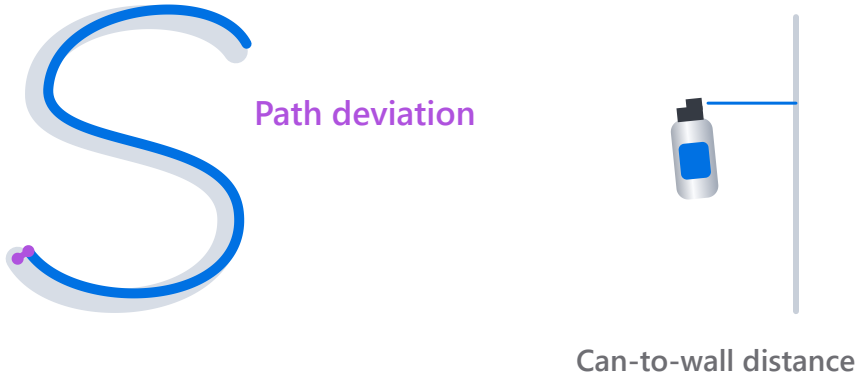
Ghost hand, tool and spatial trail

Unguided pretest

Equal-duration practice

Unguided posttest

Random assignment. Same lesson and hardware.
Small pilot: feasibility and descriptive findings.
Delayed retention test if feasible.



Painted trajectory error
Mean nearest-target distance in wall coordinates (mm)

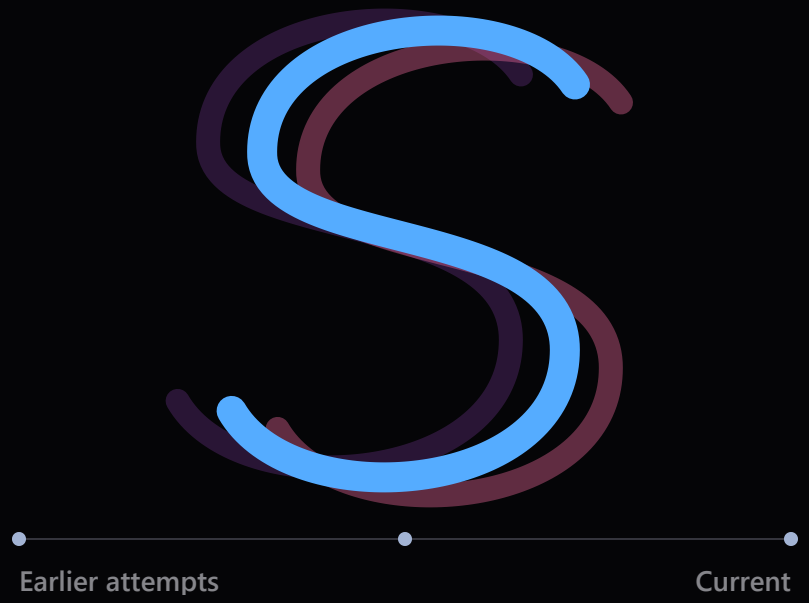
Distance stability
Variation of tool-to-wall distance while spraying (cm)

Compare pre/post change. Physical transfer requires a separate real-world task.

Levac et al., 2019. Proposed study. Example geometry, no participant results. Follow course consent requirements.

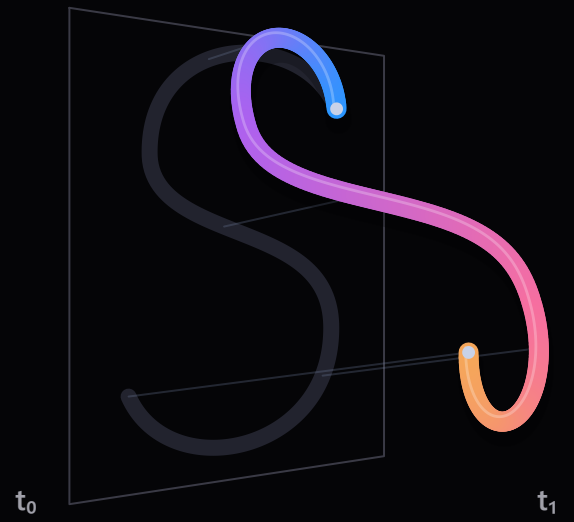
The recording becomes material

The history of making becomes part of the artwork.



Temporal Layers

Different moments.
One layered composition.



Time-to-Space

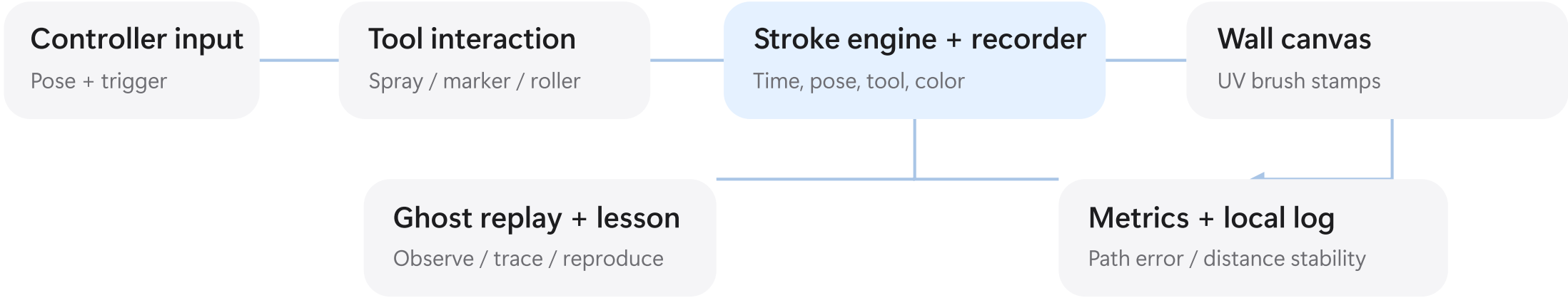
Elapsed time becomes depth.
The stroke leaves the wall.

Creative research beyond the semester MVP. Both reuse the recorded gesture.

Concept visualizations. Replay and spatial painting already exist. The proposed focus connects learning with temporal reinterpretation.

One recording. Several uses.

Meta Quest + controllers · Unity + C# · OpenXR + XR Interaction Toolkit



Paint + GPU cost

Bounded texture and brush stamps.
Profile on Quest. Lower resolution
and cap history if needed.

Replay alignment

Record in wall coordinates.
Calibrate wall height and reach.
Keep the first lesson on one plane.

Fidelity + scope

Use visual distance feedback.
Prioritize one complete spray lesson.
Keep contact tools simple.

Unity, XR input options. Proposed local architecture. No backend, fluid simulation or fast hand-tracking dependency.

One complete lesson.
A realistic first semester.

Smart Chen · Solo developer

Unity XR, interactive systems, visual design and media production.

Owner of design, implementation, creative direction, logging and evaluation.

Phase 1

Foundations

Quest input + studio.
Spray draws on the wall.

Phase 2

Tools + lesson

Basic marker + roller.
Bubble S target + lesson UI.

Phase 3

Guided practice

Record + replay.
Ghost trail + fading cues.

Phase 4

Test + polish

Unguided attempt + logs.
Usability checks + stable demo.

Semester MVP

One studio. Three basic tools. One S lesson.
Guidance, independent practice and basic remix.

When the core is stable

Small pilot. Replay scrubbing.
Temporal Layers and Time-to-Space.

Relative roadmap. Calendar dates follow the course deadline. Concept stage: no VR prototype or participant results yet.

References

Research informs the design. It does not establish outcomes for this application.

Han, Y., Ali, S. K. B. S., & Ji, L. (2022)

Use of Observational Learning to Promote Motor Skill Learning in Physical Education: A Systematic Review.
IJERPH, 19(16), 10109. 18 included studies.

Levac, D. E., Huber, M. E., & Sternad, D. (2019)

Learning and transfer of complex motor skills in virtual reality: a perspective review.
Journal of NeuroEngineering and Rehabilitation, 16, 121.

Yu, X., Lee, B., & Sedlmair, M. (2024)

Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems.
CHI 2024. DOI: 10.1145/3613904.3642143.

Wulf, G., Shea, C., & Lewthwaite, R. (2010)

Motor skill learning and performance: a review of influential factors.
Medical Education, 44(1), 75–84.

Infectious Ape. Kingspray Graffiti VR

Official PC product description on Steam.
Spray, replay, projector and interactive tutorial features.
Features can differ across platform versions.

Open Brush

Official project site.
Room-scale painting in three-dimensional space.

Krylon. How to Spray Paint

Manufacturer guidance on technique and suitable ventilation.
Used for material-practice context.

Unity. XR input options

Official Unity 6 documentation.
OpenXR, controller tracking and XR Interaction Toolkit.

Product photography: Montana Cans product photography. MTN / Spray Planet product photography. Purdy product photography.

Linked titles open original sources. Product pages and documentation checked October 27, 2026. Backup slide, outside the 7:30 talk.

Discussion

Questions to guide the next iteration

Why VR instead of a video?

Show a tracked tool's motion relative to the wall, then let the learner practice in that same space.

Why keep three tools?

They share mark and recording infrastructure. Spray gets priority. Contact tools stay basic.

How will remix be assessed?

Use reflection or artist feedback.
Trajectory error measures copying, not creativity.

Will this transfer to real paint?

Unknown. Controllers lack material feedback. A physical task would require a separate test.

What if the plan is too large?

Protect one spray lesson and useful logging. Defer the study and temporal art interactions.

What is the first thing to validate?

Can a beginner understand the ghost and trail without losing sight of their own mark?

Backup for the 3–5 minute Q&A. Physical transfer and comparative learning benefits remain open research questions.